

Foundations for
programming languages
8: Types
Answers of exercises

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Exercise

Find a term t such that

$\text{pred} (\text{succ} (\text{pred } 0)) \rightarrow t$

and construct the proof tree for that.

$$\frac{\frac{\frac{}{\text{pred } 0 \rightarrow 0} \text{ (E-PZ)}}{\text{succ} (\text{pred } 0) \rightarrow \text{succ } 0} \text{ (E-S)}}{\text{pred} (\text{succ} (\text{pred } 0)) \rightarrow \text{pred} (\text{succ } 0)} \text{ (E-P)}$$

Exercise

Find a type T such that the following type judgement holds.

if true then pred 0 else succ 0: T

Then derive the judgement by using the typing rules.

$$\frac{\frac{}{\text{true: Bool}} \text{ (T-T)} \quad \frac{\frac{}{0 : \text{Nat}} \text{ (T-Z)}}{\text{pred } 0 : \text{Nat}} \text{ (T-P)} \quad \frac{\frac{}{0 : \text{Nat}} \text{ (T-Z)}}{\text{succ } 0 : \text{Nat}} \text{ (T-S)}}{\text{if true then pred 0 else succ 0 : Nat}} \text{ (T-IF)}$$

Exercise

Find a type T such that the following type judgement holds.

$$\emptyset \vdash ((\lambda x: \text{Nat}. \text{pred } x) (\text{succ } 0)) : T$$

Then derive the judgement by using the typing rules.

$$\frac{\frac{\frac{x: \text{Nat} \in \emptyset\{x: \text{Nat}\}}{\emptyset\{x: \text{Nat}\} \vdash x : \text{Nat}} \text{ (T-VAR)}}{\emptyset\{x: \text{Nat}\} \vdash \text{pred } x : \text{Nat}} \text{ (T-P)} \quad \frac{}{\emptyset \vdash 0 : \text{Nat}} \text{ (T-Z)} \quad \frac{}{\emptyset \vdash (\lambda x: \text{Nat}. \text{pred } x) : \text{Nat} \rightarrow \text{Nat}} \text{ (T-ABS)} \quad \frac{}{\emptyset \vdash \text{succ } 0 : \text{Nat}} \text{ (T-S)} \quad \frac{\emptyset \vdash (\lambda x: \text{Nat}. \text{pred } x) : \text{Nat} \rightarrow \text{Nat} \quad \emptyset \vdash \text{succ } 0 : \text{Nat}}{\emptyset \vdash ((\lambda x: \text{Nat}. \text{pred } x) (\text{succ } 0)) : \text{Nat}} \text{ (T-APP)}$$